IFN664 Assessment 2

Technical Report

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Design and analysis of algorithms

Display the three (3) most frequently borrowed movie DVDs in descending order by the number of times the movie DVDs have been borrowed. The input of your algorithm is the hash table array where the key-item pairs are stored.

```
ALGORITHM BorrowTimeCompare(M[0..n-1])
// Compare and return the movie with the highest borrow times
//Handle the different situations for odd and even length of list
// Input: An array of n movie tuples (Title, Movie), M
// Output: The tuple (Title, Movie) with the highest borrow times

if n == 1
   return M[0]

tempMovies ← an empty list of (Title, Movie)

if n % 2 == 0
   for i ← 0 to n-1 by 2 do
      if M[i].Movie.BorrowTimes < M[i+1].Movie.BorrowTimes
         tempMovies.add(M[i+1])
      else
         tempMovies.add(M[i])
```

```
else
   tempItem ← M[0]
   for i ← 1 to n-1 by 2 do
      if M[i].Movie.BorrowTimes < M[i+1].Movie.BorrowTimes
         tempMovies.add(M[i+1])
      else
         tempMovies.add(M[i])

   if tempMovies.size > 0 and tempItem.Movie.BorrowTimes >
tempMovies[0].Movie.BorrowTimes
      tempMovies[0] ← tempItem
   else if tempMovies.size == 0
      tempMovies.add(tempItem)

if tempMovies.size == 1
   return tempMovies[0]
else
   return BorrowTimeCompare(tempMovies)
```

Algorithm Analysis:

1. The input parameter **n** characterizes the problem size, representing the number of movie tuples (Title, Movie). The computation time of the algorithm is directly impacted by the value of n.
2. The operation **M[i].Movie.BorrowTimes < M[i+1].Movie.BorrowTimes** is chosen as the basic operation of this algorithm. This compares operation is performed most frequently in the algorithm and it has the most influence on the computation time. This comparison operation will be performed numerous times throughout the algorithm.

3. The number of times the basic operation is performed remains the same in the best, worst, and average cases since every pair of movies must be compared due to the algorithm.
4. In each level of recursion, the number of comparisons is half of the current list size. Initially, which is n/2 times of comparisons. And for recursion, it will be n/4, n/8... and so on.
   So, the recurrence relation for the number of comparisons is:
   $T(n) = T(n/2) + n/2$
   At the k-th level,

   $T(n) = T(n/2^k) + n(1/2^k + 2^{k-1} + ... + 1/2)$

   $= T(n/2^k) + n(\sum_{i=1}^{k} \frac{1}{2^i})$

   $= T(n/2^k) + n(1 - \left(\frac{1}{2}\right)^k)$

   While $k = \log_2 n$:

   $T(n) = T(1) + n(1 - \frac{1}{n})$

   $= T(1) + n - 1$

   $= n - 1$

5. The efficiency of the BorrowTimeCompare algorithm is $\Theta(n)$.

Hash Table Design

For the hash table designing, a class MovieCollection has been created. In this class, there is a independent hashed function called HashedTitle(), which use library from System.Security.Cryptography to encrypt movie title into a string of decimal numbers, and before each time a movie to be add into the collection, the hashed function will trigger first, and after encrypting the title into decimal numbers, it will return to AddMovieByStaff() function. In this function, the encrypted title will be inserted into the collection, the address of this movie element will be determined while the decimal number title mod 1000, and the collision strategy of the hash table is linear probing. It will handle the collision by adding 1 to the position each time the collision happens.

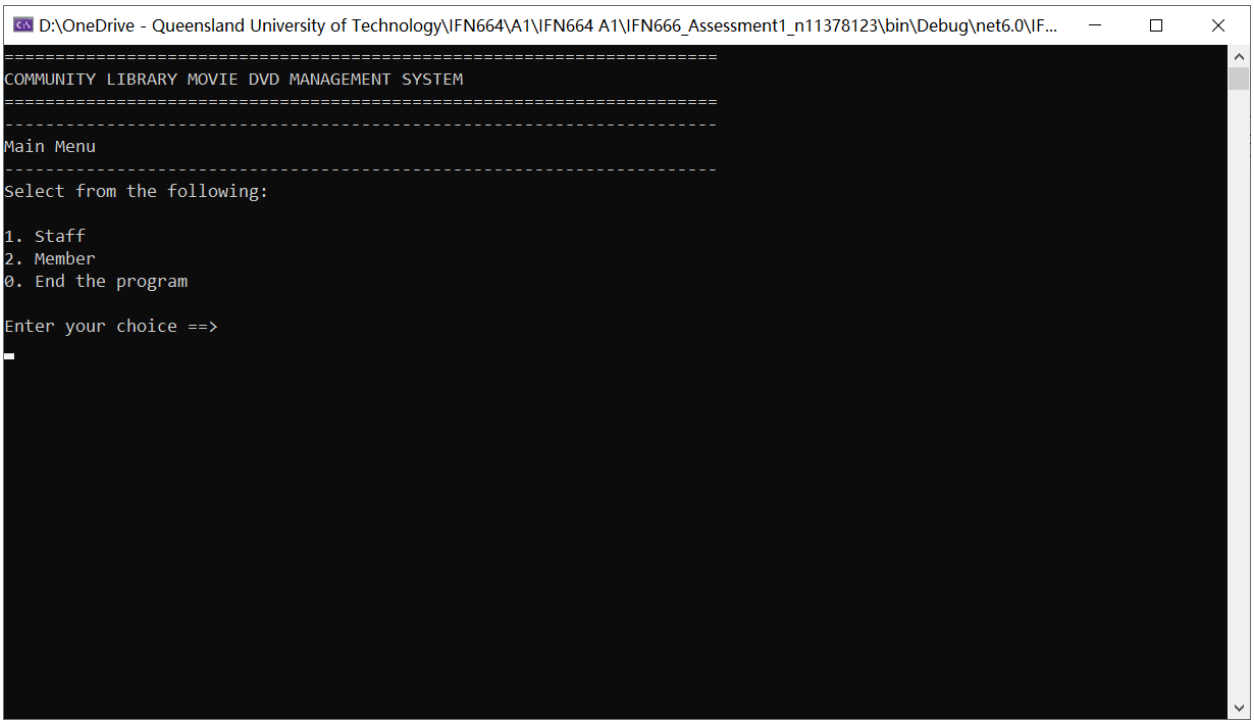
```
//Convert and mod the result to get the address for hash table
int position = (int)(hased % 1000);
if(position >= 1000)
{
    Console.WriteLine("Index out of range");
}
else
{
    //The opening address strategie Linear probing to handle the collisions and also check
    while (movies[position].Title != null)
    {
        position = (position + 1) % 1000;
    }
}
```

Test Data and Result

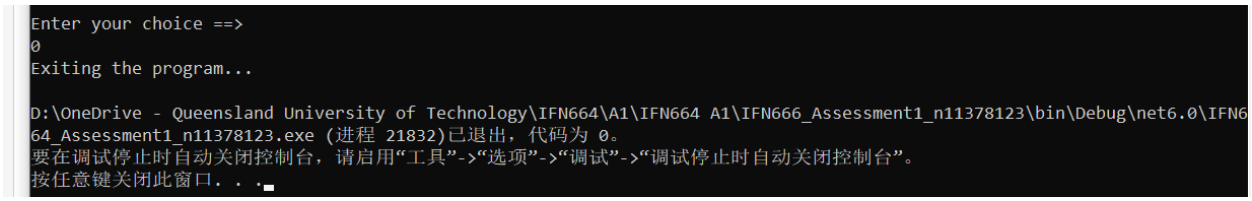
For the specialty of the program, the test data will be inserted by staff into the program, and screenshot will be put below for the data and results.

Test main menu:

The main menu will look like this, user can choose to enter a number for the function.

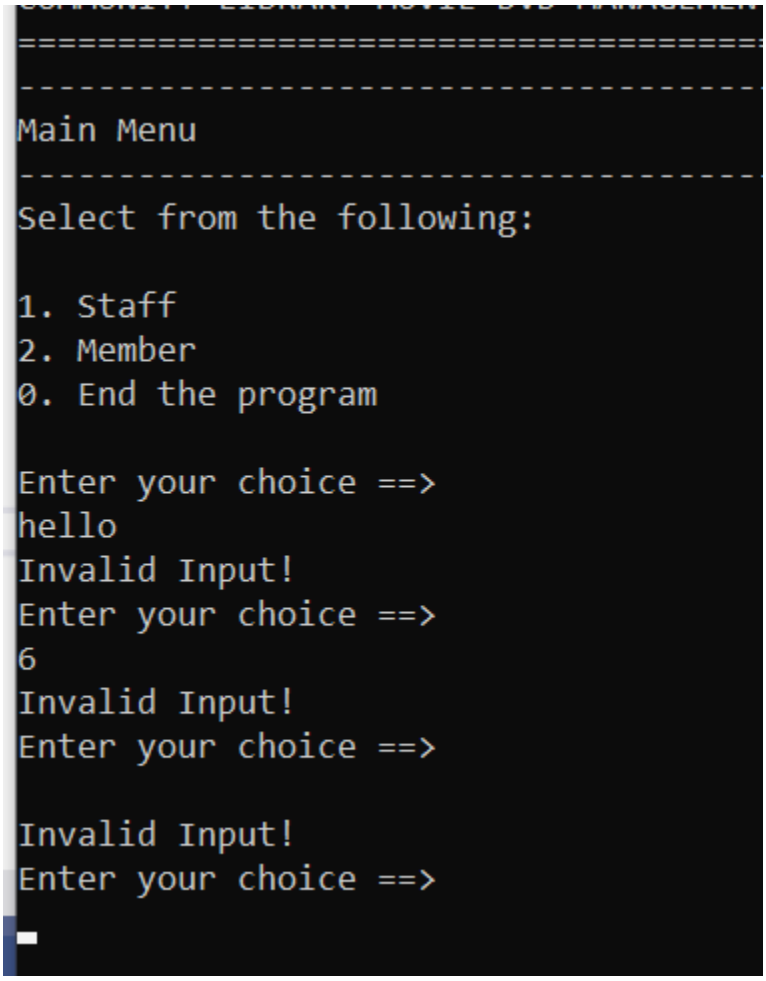


When insert 0:



Program exit.

Insert invalid input:



Input except 0, 1, 2 is all invalid with checking

Test staff login

Staff username is “staff”, password is “today123”, and with checking, only input and password match the staff menu could open:

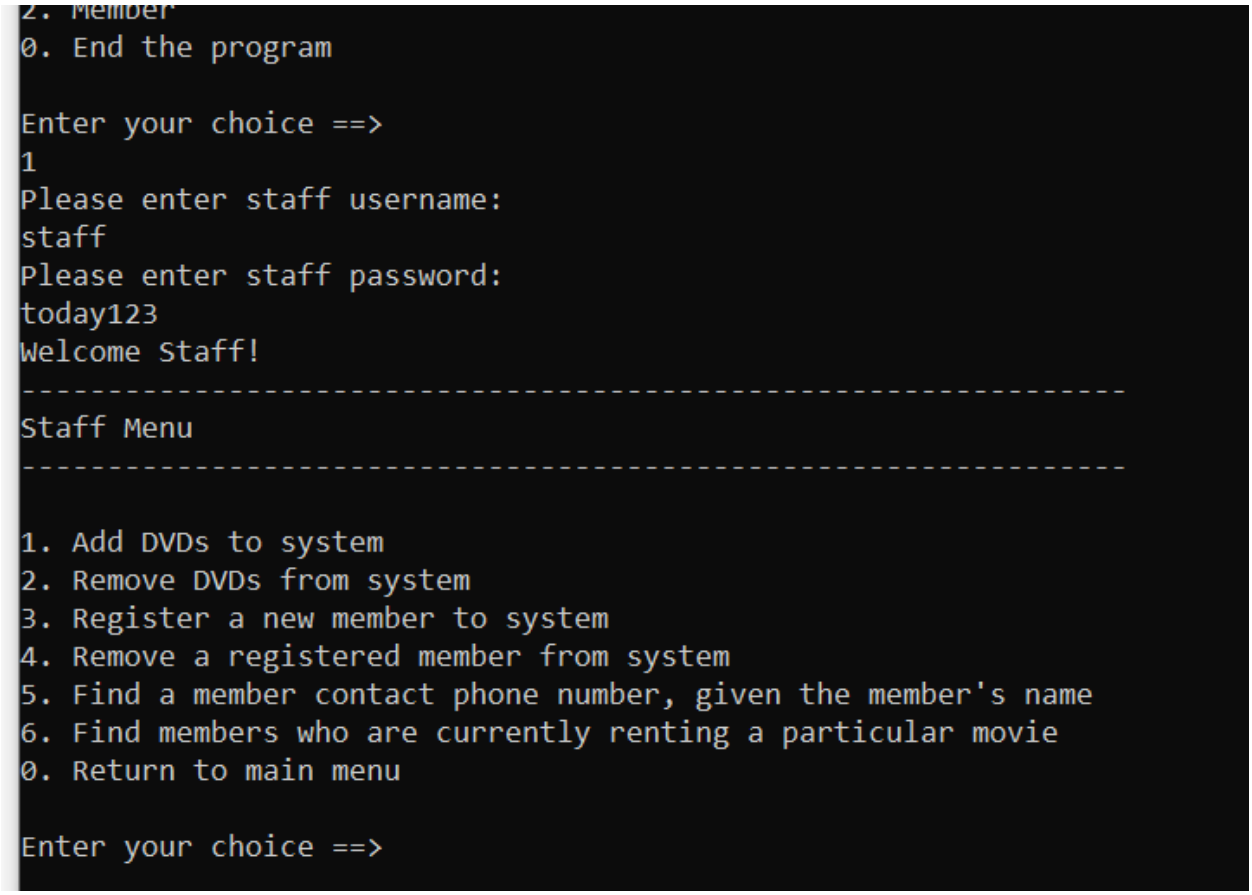
```
Enter your choice ==>
1
Please enter staff username:

Please enter staff password:

username or password not match, login failed!
---------------------------------------------------------------
Main Menu
---------------------------------------------------------------
Select from the following:
```

```
Enter your choice ==>
1
Please enter staff username:
staff
Please enter staff password:
pass
username or password not match, login failed!
---------------------------------------------------------------
Main Menu
---------------------------------------------------------------
```

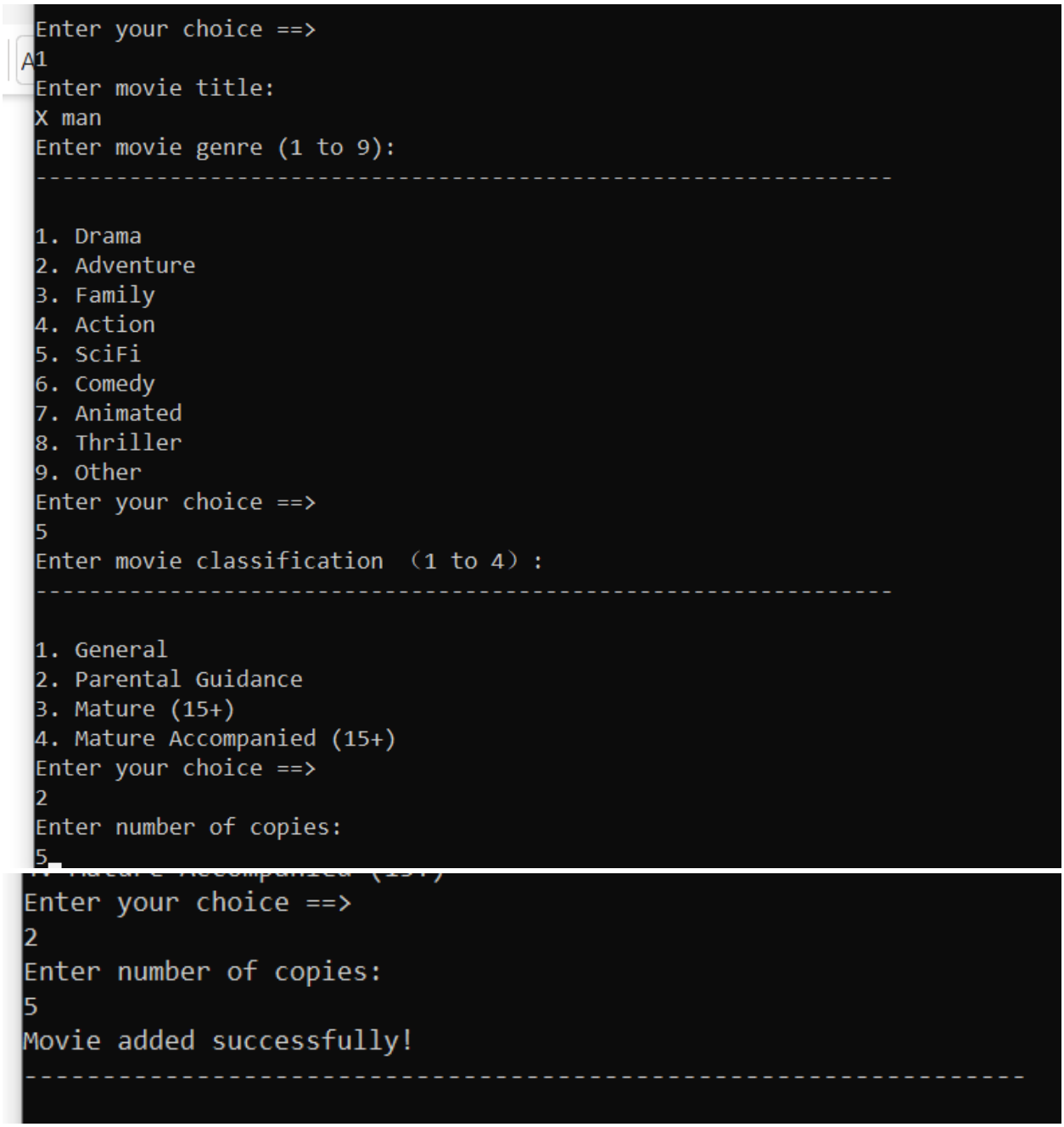
And correct staff username and password will let staff login to the staff menu:



Test add movie function

Add movie to the system will have two operations: 1. add a new movie with condition 2. add some copies to an existing movie

The check of movie if exist will prove this function:



Movie title “X man” is not contained in the community library, so it can be added. And now will add some new copies to it:

```
1. Add DVDs to system
2. Remove DVDs from system
3. Register a new member to system
4. Remove a registered member from system
5. Find a member contact phone number, given the member's name
6. Find members who are currently renting a particular movie
0. Return to main menu

Enter your choice ==>
1
Enter movie title:
X man
Add new copies into movie: X man, please enter a number:
5
Copies added successfully
----------------------------------------------------------------
```

We can add another movie to test some input validation:

Movie title cannot be empty

```
Enter your choice ==>
1
Enter movie title:

Title cannot be empty. Please enter a valid title.
Enter movie title:
```

The movie genre should be chosen from 1 to 9:

```
9. Other
Enter your choice ==>
ada
Invalid input, please enter again
------------------------------------------------------------------


1. Drama
2. Adventure
3. Family
4. Action
5. SciFi
6. Comedy
7. Animated
8. Thriller
9. Other
Enter your choice ==>
10
Invalid input, please enter again
------------------------------------------------------------------
```

Same validation for classification from 1 to 4:

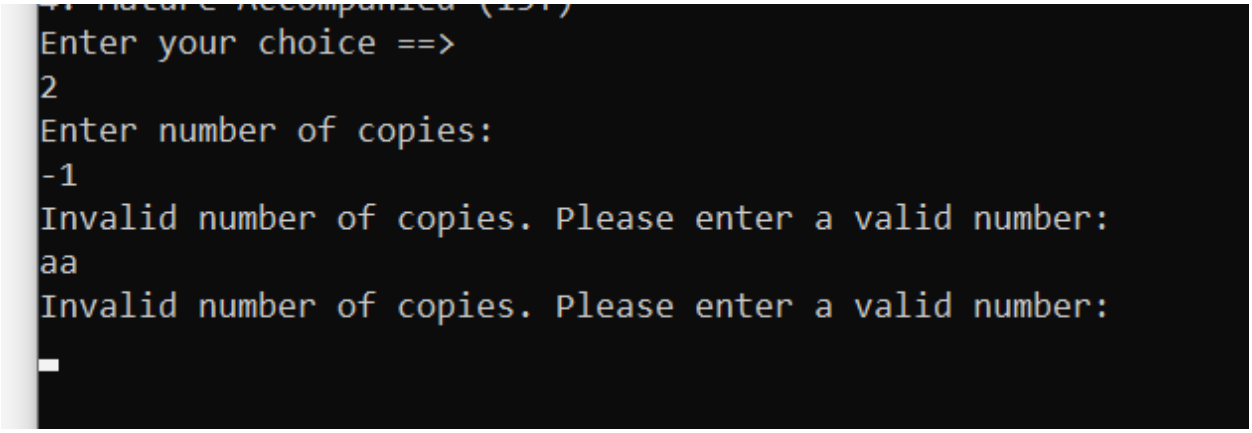
```
Enter movie classification  (1 to 4) :
------------------------------------------------------------------


1. General
2. Parental Guidance
3. Mature (15+)
4. Mature Accompanied (15+)
Enter your choice ==>
as
Invalid input, please enter again
------------------------------------------------------------------


1. General
2. Parental Guidance
3. Mature (15+)
4. Mature Accompanied (15+)
Enter your choice ==>
6
Invalid input, please enter again
------------------------------------------------------------------

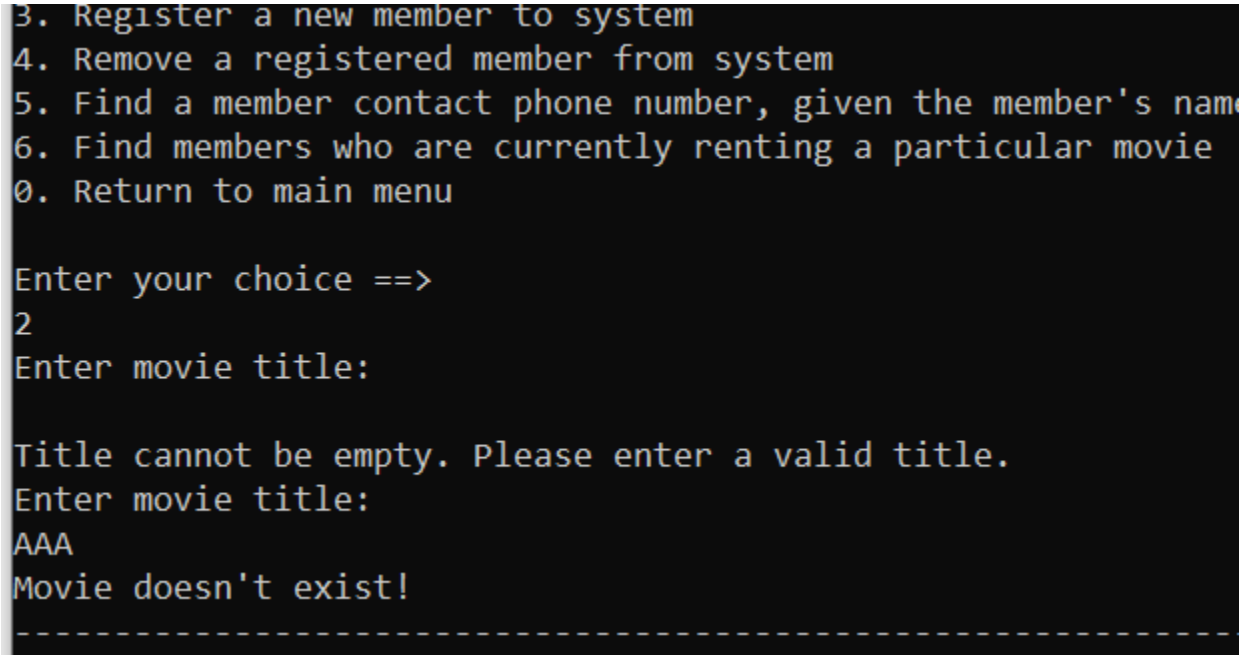

1. General
2. Parental Guidance
```

And number of copies can only be numbers equal or larger than 0:

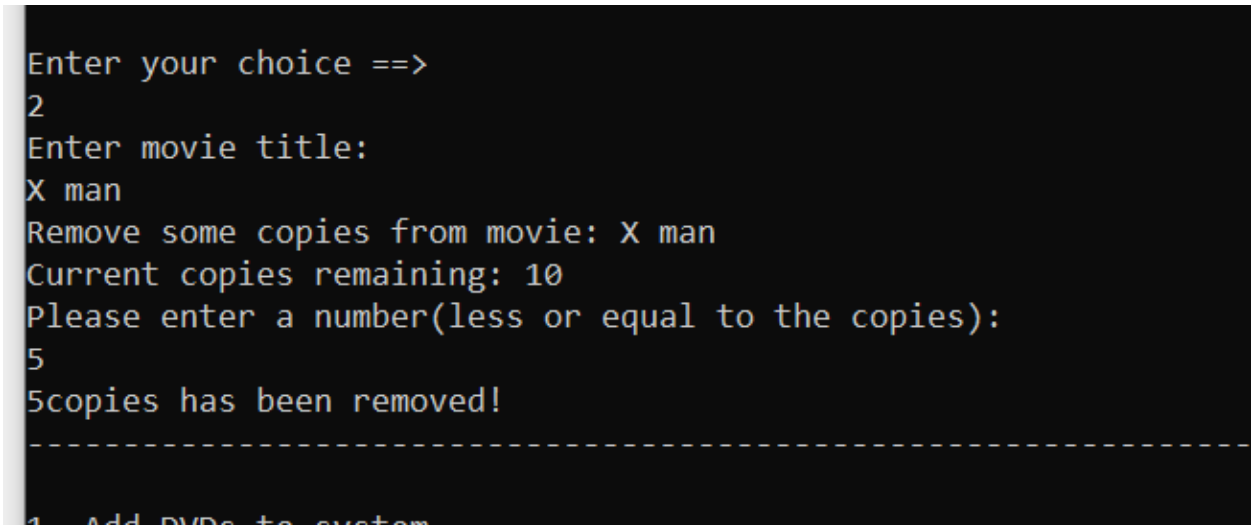


Test remove movie function

To remove a movie, movie title will be required, and the system will check if the movie exist or not, and the movie title should not be empty (as well as add movie):

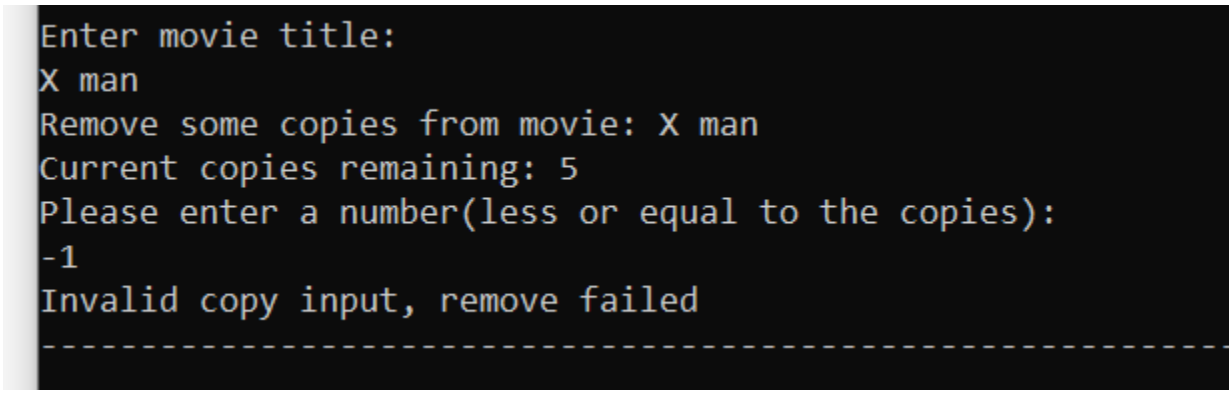


Now test remove some DVDs from X man:

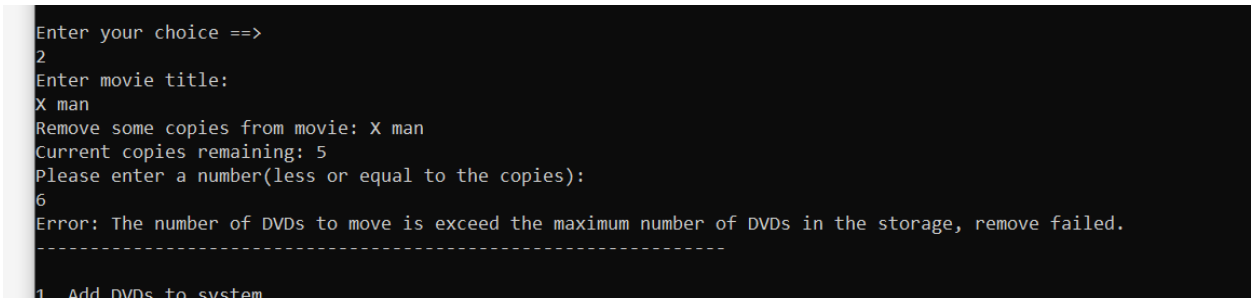


5 copies have been removed successfully.

And this copies also have validation check:



There are 5 DVDs in X man remaining, if remove more than 5 DVDs, it will send error messages:



And if remove all DVDs for this movie, the movie will be removed:

```
Enter your choice ==>
2
Enter movie title:
X man
Remove some copies from movie: X man
Current copies remaining: 5
Please enter a number(less or equal to the copies):
5
Movie has been removed.
------------------------------------------------------------
```

Test member register

Staff can register new members to system, and there is validation check for phone number (10 digits for Australia style), and password (4 digits number)

```
Enter member firstname:
Tom
Enter member lastname:
Bruce
Member Tom Bruce
Enter phone number (10 digits): 1
Invalid phone number. Please try again.
Member Tom Bruce
Enter phone number (10 digits): a
Invalid phone number. Please try again.
Member Tom Bruce
Enter phone number (10 digits):
```

```
Enter phone number (10 digits): 1234567890
Phone number is valid.
```

```
Set the password for member (4 digits): asd6
Invalid password. Please try again.
Set the password for member (4 digits): 4
Invalid password. Please try again.
Set the password for member (4 digits): 456123
Invalid password. Please try again.
Set the password for member (4 digits):
```

```
Set the password for member (4 digits): 6666
Password is Valid.
Member added: Tom Bruce
----------------------------------------------------------------

First name: Tom
Last name: Bruce
Phone number: 1234567890
Password: 6666
----------------------------------------------------------------
```

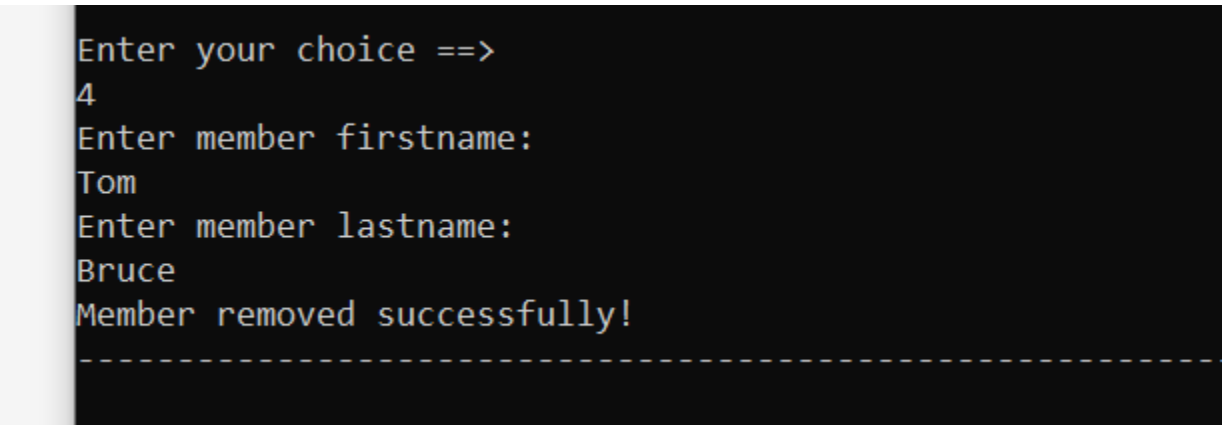
And a member has been successfully registered.

There will be checks if the user has already existed:

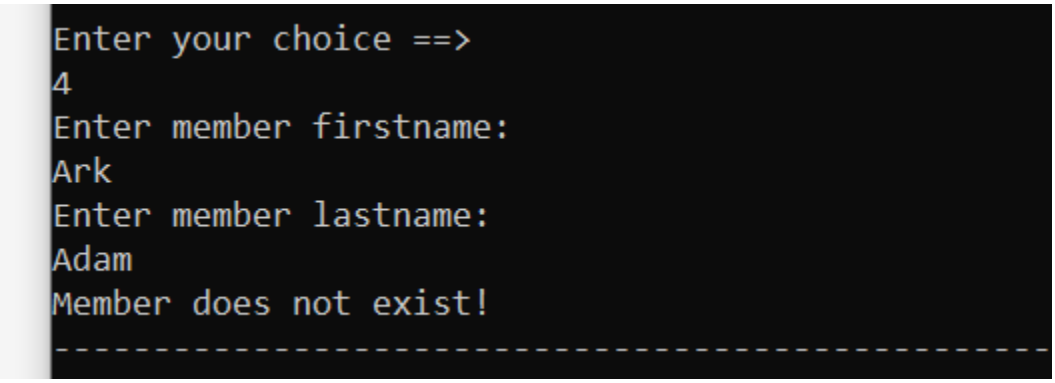
```
Enter your choice ==>
3
Enter member firstname:
Tom
Enter member lastname:
Bruce
Member Tom Bruce
Enter phone number (10 digits): 5555555555
Phone number is valid.
Set the password for member (4 digits): 6666
Password is Valid.
Member already exists.
----------------------------------------------------------
```

Test remove member function

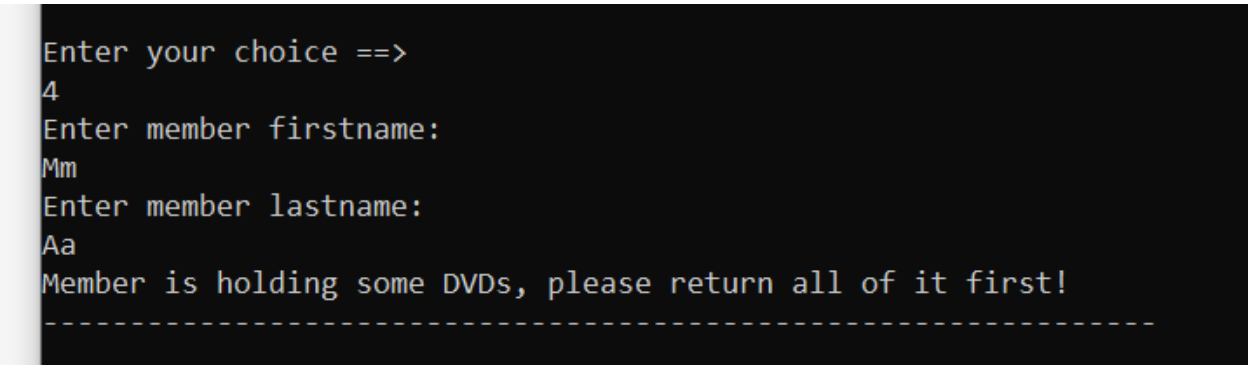
Remove member needs first name and last name:



If a member does not exist:



And will show if member still hold some DVDs, it cannot be deleted:



Test find member phone number:

This also check if member exists:

```
Enter your choice ==>
5
Enter member firstname:
Tom
Enter member lastname:
Bruce
Member does not exist!
-----------------------------------------------------
```

And it will show phone number for an exist user:

```
Enter your choice ==>
5
Enter member firstname:
Mm
Enter member lastname:
Aa
Member name: Mm Aa
Phone number: 1234567890
-----------------------------------------
```

List member who borrowed particular movies:

First, we register some members and movies:

```
Password is Valid.
Member added: Andy Fan
-------------------------------------------------------------

First name: Andy
Last name: Fan
Phone number: 7894561231
Password: 6666
-------------------------------------------------------------
```

```
------------------------------------------------------------------

First name: Tom
Last name: Bruce
Phone number: 4561237890
Password: 6666
------------------------------------------------------------------
```

```
------------------------------------------------------------------

Movie title: AA
Movie genre: Adventure
Movie classification: G
Movie copies: 5
Movie borrow times: 0
------------------------------------------------------------------

Movie title: Avenagers
Movie genre: Action
Movie classification: M15Plus
Movie copies: 9
Movie borrow times: 0
------------------------------------------------------------------

Movie title: Mr Bean
Movie genre: Comedy
Movie classification: G
Movie copies: 8
Movie borrow times: 0
------------------------------------------------------------------

Movie title: The Matrix
Movie genre: SciFi
Movie classification: PG
Movie copies: 5
Movie borrow times: 1
------------------------------------------------------------------
```

```
------------------------------------------------------------------

Movie title: X man
Movie genre: SciFi
Movie classification: M15Plus
Movie copies: 6
Movie borrow times: 0
------------------------------------------------------------------
```

Let Tom Bruce and Andy Fan borrow X man:

```
Enter your choice ==>
3
Enter movie title:
X man
Borrow movie: X man
Movie Borrowed!
------------------------------------------------
```

And list all members borrowed X man:

```
Enter your choice ==>
6
Enter movie title:
X man
Members currently renting X manare:
Tom Bruce
Andy Fan
------------------------------------------------------------
```

Test return to main menu:

Select function 0 to 6 also have validations:

```
------------------------------------------------------------------


1. Add DVDs to system
2. Remove DVDs from system
3. Register a new member to system
4. Remove a registered member from system
5. Find a member contact phone number, given the member's name
6. Find members who are currently renting a particular movie
0. Return to main menu

Enter your choice ==>
a
Invalid Input!
------------------------------------------------------------------
```

And case 0 will back to the main menu:

```
Enter your choice ==>
0
Back to main menu
------------------------------------------------------------------------
Main Menu
------------------------------------------------------------------------
Select from the following:

1. Staff
2. Member
0. End the program

Enter your choice ==>
```

Test member login:

Member login will ask firstname, lastname and password, and the validation is all field cannot be empty, check if member exist and check if password matches:

```
Enter your choice ==>
2
Enter member firstname:

Title cannot be empty. Please enter a valid title.
Enter member firstname:
ad
Enter member lastname:
23
Invalid index.
Member does not exist!
------------------------------------------------------------------------
Main Menu
```

```
Enter your choice ==>
2
Enter member firstname:
Andy
Enter member lastname:
Fan
Member Andy FanPlease enter your password(4 digits):
12345
Invalid password!
------------------------------------------------------------------------
```

If member name and password matches, member can login and see member menu:

```
Enter your choice ==>
2
Enter member firstname:
Andy
Enter member lastname:
Fan
Member Andy FanPlease enter your password(4 digits):
6666
Password matches, welcome, member Andy Fan
---------------------------------------------------------------
Member Menu
---------------------------------------------------------------

1. Browse all the movies
2. Display all the information about a movie, given the title of the movie
3. Borrow a movie DVD
4. Return a movie DVD
5. List current borrowing movies
6. Display the top 3 movies rented by the members
0. Return to main menu

Enter your choice ==>
```

Test browse all movies function

This menu also includes validations, only can choose from 0 to 6:

```
Enter your choice ==>
ada
---------------------------------------------------------------

1. Browse all the movies
2. Display all the information about a movie, given the title of the movie
3. Borrow a movie DVD
4. Return a movie DVD
5. List current borrowing movies
6. Display the top 3 movies rented by the members
0. Return to main menu

Enter your choice ==>
```

And case 1 can list all movies in stock:

```
Enter your choice ==>
1
------------------------------------------------------------------

Movie title: AA
Movie genre: Adventure
Movie classification: G
Movie copies: 5
Movie borrow times: 0
------------------------------------------------------------------

Movie title: Avenagers
Movie genre: Action
Movie classification: M15Plus
Movie copies: 9
Movie borrow times: 0
------------------------------------------------------------------

Movie title: Mr Bean
Movie genre: Comedy
Movie classification: G
Movie copies: 8
Movie borrow times: 0
------------------------------------------------------------------

Movie title: The Matrix
Movie genre: SciFi
Movie classification: PG
```

```
------------------------------------------------------------------

Movie title: The Matrix
Movie genre: SciFi
Movie classification: PG
Movie copies: 5
Movie borrow times: 1
------------------------------------------------------------------

Movie title: X man
Movie genre: SciFi
Movie classification: M15Plus
Movie copies: 4
Movie borrow times: 2
------------------------------------------------------------------

1. Browse all the movies
2. Display all the information about a movie, given the title of the movie
3. Borrow a movie DVD
4. Return a movie DVD
5. List current borrowing movies
```

Test find movie by title function:

A movie can be found with its title, validation check for not empty and if movie exist:

```
Enter your choice ==>
2
Enter movie title:

Title cannot be empty. Please enter a valid title.
Enter movie title:
A man
Movie doesn't exist!
------------------------------------------------------------------
```

If movie in the records it will show detail of this movie:

```
Enter your choice ==>
2
Enter movie title:
Mr Bean
------------------------------------------------------------------

Movie title: Mr Bean
Movie genre: Comedy
Movie classification: G
Movie copies: 8
Movie borrow times: 0
------------------------------------------------------------------

1. Browse all the movies
2. Display all the information about a movie, given the title of the movie
```

Test Borrow movie function

To borrow a movie, movie title should be provided, same validation checks as find movie function:

```
Enter your choice ==>
3
Enter movie title:

Title cannot be empty. Please enter a valid title.
Enter movie title:
A man
Movie does not exist!
------------------------------------------------------------------
```

And if a movie matches, member can borrow it:

```
Enter your choice ==>
3
Enter movie title:
Mr Bean
Borrow movie: Mr Bean
Movie Borrowed!
----------------------------------------------------------------
```

And member can only borrow one copy of the same movie:

```
Enter your choice ==>
3
Enter movie title:
Mr Bean
Movie has been borrowed, can not borrow the same movie twice!
----------------------------------------------------------------
```

One member can borrow maximum 5 movies, more is not allowed:

We first added another movie into the library to make sure there are more than 5 for member to borrow:

```
Movie title: The Flash
Movie genre: Adventure
Movie classification: M15Plus
Movie copies: 6
Movie borrow times: 0
----------------------------------------------------------------
```

This user is currently borrowing 5 movies:

```
5
Member Andy Fan
Here are the movies you have borrowed:
----------------------------------------
1. X man
2. The Matrix
3. Mr Bean
4. AA
5. The Flash
------------------------------------------------------------------
```

And if we try to borrow another one:

```
Enter your choice ==>
3
Enter movie title:
Avenagers
Borrow movie: Avenagers
You have already borrowed five movies, borrow failed!
------------------------------------------------------------
```

It will show warnings and borrow failed.

Test return function:

Member can return a movie to library, also validation checks:

```
Enter your choice ==>
4
Enter movie title:

Title cannot be empty. Please enter a valid title.
Enter movie title:
A man
Movie does not exist!
------------------------------------------------------------------
```

If movie is not borrowed by member, it will show the warning:

```
Enter your choice ==>
4
Enter movie title:
The Matrix
Your have no borrowed record on this movie!
------------------------------------------------------------
```

If member have borrowed the movie which matches the movie in the library, it will return movie:

```
Enter your choice ==>
4
Enter movie title:
Mr Bean
Return your DVD!
Movie return successful!
------------------------------------------------------------
```

Test List current borrowed movie function

List the current movie a member borrowed.

```
Enter your choice ==>
5
Member Andy Fan
Here are the movies you have borrowed:
---------------------------------------
1. X man
2. The Matrix
3.
4. AA
5. The Flash
------------------------------------------------------------
```

This user has borrowed 5 movies and returned movie “Mr Bean” back, so it will list what is currently he borrowed.

Test list three movies with largest borrow times:

Before we test this, we need to create some more members to borrow the movies so it could have differences between borrow times:

```
Enter member firstname:
Lame
Enter member lastname:
Ray
Member Lame Ray
Enter phone number (10 digits): 666
Invalid phone number. Please try again.
Member Lame Ray
Enter phone number (10 digits): 6666
Invalid phone number. Please try again.
Member Lame Ray
Enter phone number (10 digits): 1234567890
Phone number is valid.
Set the password for member (4 digits): 6666
Password is Valid.
Member added: Lame Ray
------------------------------------------------------------
```

```
Enter your choice ==>
3
Enter member firstname:
Alex
Enter member lastname:
Mask
Member Alex Mask
Enter phone number (10 digits): 4578912303
Phone number is valid.
Set the password for member (4 digits): 6
Invalid password. Please try again.
Set the password for member (4 digits): 6666
Password is Valid.
Member added: Alex Mask
------------------------------------------------------------
```

After borrowing some movies by these five members, the total list of movies information is look like this:

```
Enter your choice ==>
1
-----------------------------------------------------------------

Movie title: AA
Movie genre: Adventure
Movie classification: G
Movie copies: 3
Movie borrow times: 2
-----------------------------------------------------------------

Movie title: Avenagers
Movie genre: Action
Movie classification: M15Plus
Movie copies: 6
Movie borrow times: 3
-----------------------------------------------------------------

Movie title: Mr Bean
Movie genre: Comedy
Movie classification: G
Movie copies: 7
Movie borrow times: 3
-----------------------------------------------------------------

Movie title: The Flash
Movie genre: Adventure
Movie classification: M15Plus
Movie copies: 4
```

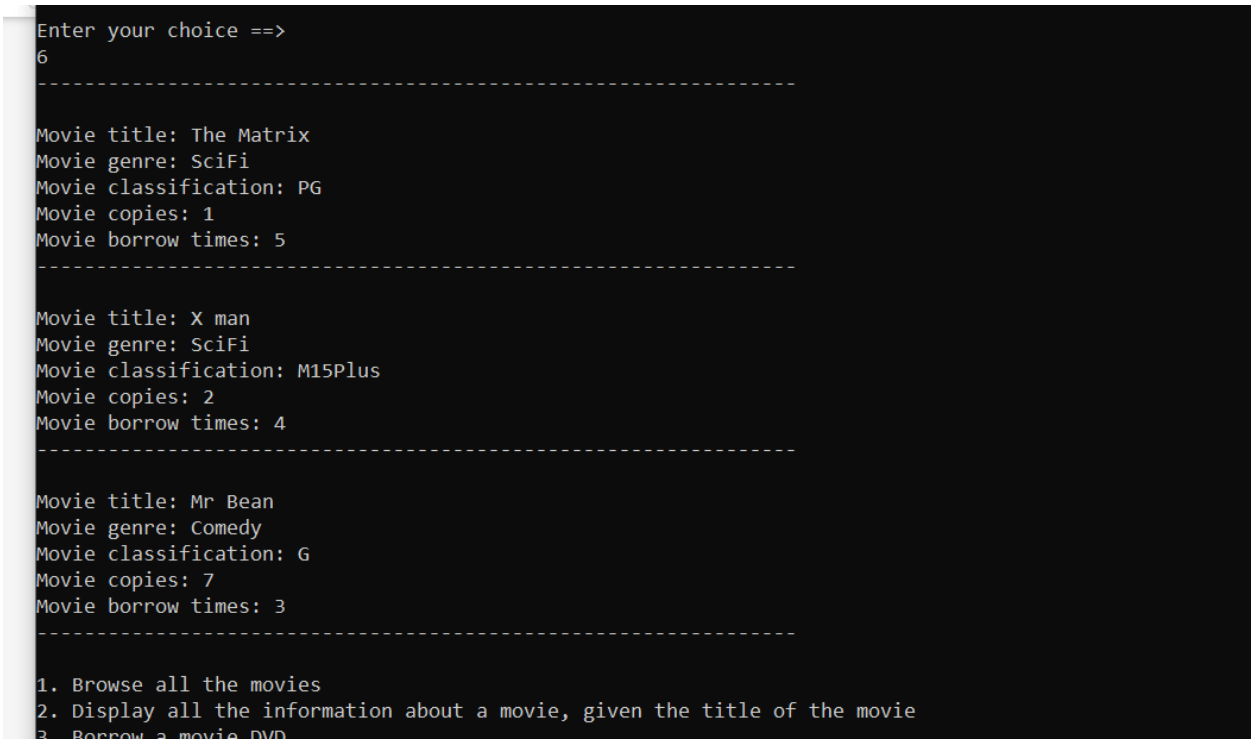
```
Movie title: The Flash
Movie genre: Adventure
Movie classification: M15Plus
Movie copies: 4
Movie borrow times: 2
-----------------------------------------------------------------

Movie title: The Matrix
Movie genre: SciFi
Movie classification: PG
Movie copies: 1
Movie borrow times: 5
-----------------------------------------------------------------

Movie title: X man
Movie genre: SciFi
Movie classification: M15Plus
Movie copies: 2
Movie borrow times: 4
-----------------------------------------------------------------

1. Browse all the movies
```

And we will run the function to show 3 most popular movies:



It shows three movies with the most borrowed which matches the result which is shown on the total list.